WEEK-4

Cumulative Distribution Function (CDF)

$$F_X(x) = P(X \le x)$$

→ key properties:

- $0 \le F(x) \le 1$
- Non-Decreasing
- $F(-\infty) = 0, \; F(\infty) = 1$
- $P(a < X \le b) = F_X(b) - F_X(a)$

To find range of X look where CDF jumps:

Q① The cumulative distribution func. of a discrete random var. X is given by:

$$F(x) = \begin{cases} 0 & \text{if } x < -2 \\ 0.15 & \text{if } -2 \le x < -1 \\ 0.35 & \text{if } -1 \le x < 1 \\ 0.75 & \text{if } 1 \le x < 2 \\ 1 & \text{if } x \ge 2 \end{cases}$$

Sol.

$x = -2$
$Y = 2(-2)^2 + 1 \Rightarrow 9$

$x = -1$
$Y = 2(-1)^2 + 1 \Rightarrow 3$

$X = 1$
$Y = 2(1)^2 + 1 \Rightarrow 3$

$X = 2$
$Y = 2(2)^2 + 1 \Rightarrow 9$

$P(Y=3) = P(X=-1) + P(X=1)$
$\Rightarrow (0.35 - 0.15) + (0.75 - 0.35)$
$\Rightarrow 0.60$

$P(Y=9) = P(X=-2) + P(X=2)$
$\Rightarrow (0.15 - 0) + (1 - 0.75)$
$\Rightarrow 0.40$

Q (2) Let a discrete random var, X have a CDF F_X. Assume that $Y = \frac{X-\mu}{\sigma}$ where μ and σ are the means and the S.D of X, respectively. If F_Y is the distribution func. of Y, which of the following Option is true?

Sol.

$$F_Y(y) = P(Y \le y)$$
$$= P\left(\frac{X-\mu}{\sigma} \le y\right)$$
$$= P(X-\mu \le \sigma y)$$
$$= \boxed{P(X \le \sigma y + \mu)} \approx F_X(\mu + \sigma y)$$

$$F_Y(y) = F_X(\mu + \sigma y)$$

Q (3) Let $X \sim$ Geometric (p). Find the CDF of x

Ans. $\Rightarrow$ $F(x) = \begin{cases} 0 & x < 1 \\ 1-(1-p)^k & k \le x < k+1 \end{cases}$ $\quad k = 1, 2, 3, \ldots \infty$

Sol.

$$f(x) = (1-p)^{x-1} \cdot p$$
$$f(k) = (1-p)^{k-1} \cdot p$$
$$f(X=k) = P(X \le k) = P(X=1) + P(X=2) + P(X=3) + \ldots + P(X=k)$$
$$\Rightarrow p + (1-p)^{2-1} \cdot p + (1-p)^{3-1} \cdot p + \ldots + (1-p)^{k-1} \cdot p$$
$$\Rightarrow \frac{p(1-r^n)}{(1-r)} \Rightarrow \frac{p(1-(1-p)^n)}{1-(1-p)}$$
$$\Rightarrow \boxed{1-(1-p)^n}$$

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8. ☆ For Graph based Ques see the CDF jumps of main X-coordinate to find Probability distrib.

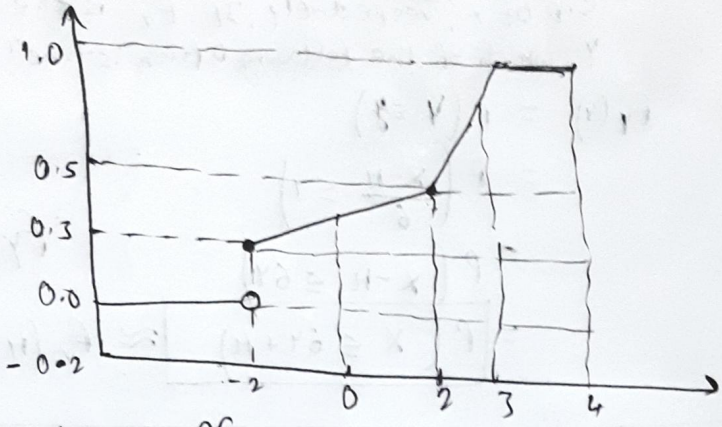
Continuous Random Variable

$P(X=a) = 0$ for any single value of a.

- No Jumps
- $P(X=x) = 0$

$P(a < X < b) = F(b) - F(a)$

Q. Consider the following CDF of a random variable X to answer the question (6), (7) and (8).



Q. Find the value of $P(X=-2)$.

$\hookrightarrow P(X=-2) = F(-2) - F(-2^-)$

$0.3 - 0 = 0.3$

Q. Find the value of $P(0 \le X \le 2.5)$.

Sol. $P(a \le X \le b) = F(b) - F(a^-)$

$\Rightarrow P(0 \le X \le 2.5) = F(2.5) - F(0)$

Step:1 Find eqn. of line (-2 to 2)

$$m = \frac{0.5 - 0.3}{2-(-2)} = 0.05$$

$$y - y_1 = m(x - x_1)$$

$$y - 0.3 = 0.05(x+2)$$

$$\boxed{y = 0.05x + 0.4}$$

$$F(0) = 0.5(0) + 0.4 = 0.4$$

Between 2 and 3 $\rightarrow$ $F(x) = 0.5x - 0.5$

$F(2.5) = 0.75$

$$P(0 \le X < 2.5) = 0.75 - 0.4 = 0.35$$

Q. Find $P(X \ge 1)$

$\hookrightarrow P(X \ge 1) = 1 - P(X < 1) = 1 - F(1) \Rightarrow 0.55$

$0.05(1) + 0.4 = 0.45$

Probability Density Func. (PDF)

A continuous random variable x with CDF $f_x(x)$ is said to have a PDF $f_x(x)$ if, for all x_0

$$f_x(x_0) = \int_{-\infty}^{x_0} f_x(x)\,dx = 1$$

- CDF is the integral of PDF $\rightsquigarrow \frac{d}{dx}(\text{CDF}) = \text{PDF}$

$\uparrow$ PDF $\rightarrow$ higher chance that x lies there

- PDF (Properties) $\rightarrow$ $f(x) \geq 0$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$f(x)$ is piecewise continuous

Q.① The PDF of rand. var. x is given as:

$$f_x(x) = \begin{cases} 0.1 & 0 \leq x < 1 \\ Cx & 1 \leq x < 2 \\ 0.3 & 2 \leq x < 3 \\ 0 & \text{otherwise} \end{cases}$$

find the value of $P(1 < x < 2.5)$

Sol.

$$\int_0^1 (0.1)\,dx + \int_1^2 Cx\,dx + \int_2^3 (0.3)\,dx = 1$$

$$C = 0.4$$

$$P(1 < x < 2.5) = \int_1^2 \overset{\nearrow 0.4}{C}x\,dx + \int_2^{2.5} (0.3)\,dx$$

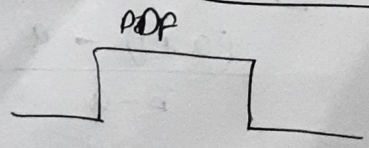
$$= 0.75$$

(Uniform Distribution) $x \sim \text{Uniform}[a,b]$

Standard Distributions

- PDF

$$f_x(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$



- CDF

$$F_x(x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x \geq b \end{cases} \star$$

(Exponential Distribution) $X \sim Exp(\lambda)$ $\lambda = \frac{1}{\mu}$

- PDF $(\lambda e^{-\lambda x})$

$$f_X(x) = \begin{cases} \lambda \exp(-\lambda x) & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

- CDF

$$F_X(x) = \begin{cases} 0 & x \le 0 \\ 1 - \exp(-\lambda x) & x > 0 \end{cases}$$

$(1 - e^{-\lambda x})$

(Normal Distribution) $X \sim Normal(\mu, \sigma^2)$ → mode ≈ mean

- PDF

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$$

- CDF

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$$

o Standard normal: Norma $(0,1)$

at μ x will most likely to occur

Q.① Let $X \sim$ Uniform $[2,6]$. Find the value of $P(1<x<5)$.

Sol. $[2,6]$ → $a=2$, → $b=6$

$$f(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \le x < b \\ 1 & x > 6 \end{cases}$$

$P(1<x<5)$

→ $F(5) - F(1)$

$\frac{5-2}{6-2} - 0 = \frac{3}{4} \simeq \boxed{0.75}$

Q. Let $X \sim$ Uniform $[a,b]$. If $P(X \le a+2) = 0.2$, What will be the value of $P(X \ge b-2)$?

Sol. $P(X \le a+2) = 0.2$

$\frac{(a+2) - a}{b-a} = 0.2$

$\frac{2}{b-a} = \frac{2}{10}$

$b - a = 10$

$P(X \ge b-2) = 1 - P(X \le b-2)$

$\Rightarrow 1 - F_X(b-2)$

$\Rightarrow 1 - \left[\frac{(b-2)-a}{b-a}\right]$

$\Rightarrow 1 - \left[\frac{-2+10}{10}\right]$

Ans = 0.2

Q. (?) The amt. of milk produced every day by a dairy is uniformly between 100 lt and 120 lt. What is the prob. that the amt. of milk produced is more than 115 lt given that on that day, more than 110 lt of milk was produced?

Sol. $P(X>115 \mid X>110) = \dfrac{P(X>115) \text{ and } P(X>110)}{P(X>110)}$

$X \sim uni\,[100, 120]$

$\Rightarrow \dfrac{P(X>115)}{P(X>110)} \Rightarrow \dfrac{1-P(X\le 115)}{1-P(X\le 110)}$

$\Rightarrow \dfrac{1-F(115)}{1-F(110)} = \dfrac{1-\frac{115-100}{120-100}}{1-\frac{110-100}{120-100}} = \dfrac{5}{10} = 0.5$

Q. Let $X \sim Exp(5)$. Find the value of $P(X \le 10)$.

Sol. $X \sim Exp(\lambda)$

$\hookrightarrow 1-e^{-\lambda x} = 1-e^{-5(10)} = 1-e^{-50}$

Q. Let $X \sim Exp(2)$. Find the value of $P(3 \le X \le 5)$

Sol. $P(3 \le x \le 5) = F(5) - F(3)$

$(1-e^{-2(5)}) - (1-e^{-(2)(3)})$

$\Rightarrow \boxed{e^{-6} - e^{-10}}$

Q. $\boxed{\text{Z-score}}$

Let $X \sim Normal(\overset{\mu}{50}, \overset{\sigma^2}{25})$. Find $P(X \ge 50)$.

Sol. $\boxed{Z = \dfrac{X-\mu}{\sigma}} = \dfrac{(50-50)}{5} \Rightarrow 0$

Z-score → how for a value is from mean

$Z=0 \rightarrow$ exactly at mean

$Z>0 \rightarrow$ above the mean

$Z<0 \rightarrow$ below the mean

Use standard normal property:

$\boxed{P(Z \ge 0) = 0.5}$

- It is perfectly symmetric
- Half area is on each side.

$P(X \ge 50) = P(Z<0)$

$\Rightarrow 0.5$

Q. (2) Let $X \sim \text{Normal}(50, 100)$. find $P(45 < X < 62)$.
(Assume that F_Z denote the CDF of Standard nomal Distrib)

Sol.

Z-Score for 45 $\Rightarrow Z = \frac{X - \mu}{\sigma} \Rightarrow \frac{45 - 50}{10} = \frac{-5}{10} = -0.5$

Z score for 62 $\Rightarrow Z = \frac{62 - 50}{10} = 1.2$

$$P(45 < X < 62) = P\left((-0.5)_Z < X < (1.2)_Z\right)$$

$$\boxed{F_Z(1.2) - F_Z(-0.5)}$$

WEEK-5

Monotonic → consistently increasing or decreasing
differentiable → diffentiable at point → if der